

Kaitlyn Dickinson

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Skills

Languages: Python, SQL, JavaScript, Java, C++

Frameworks & Libraries: Django, FastAPI, Flask, LangChain, Angular, Streamlit

Cloud & DevOps: AWS (EC2, SQS, S3), Apache Airflow, Docker, CI/CD, Nginx

Databases & Tools: Snowflake, ChromaDB, PostgreSQL, Databricks, Git, Jira

Experience

EchoStar

Englewood, CO

Application Developer III

12/2024 - Present

- Migrated 7+ business-critical scripts to Apache Airflow DAGs: introducing scheduling, dependency management and automated recovery where none existed and saving the team manual reconciliation.
- Built a pipeline that ingests and tracks changes in knowledge base articles, filters by metadata to isolate content and syncs to a customer service platform, exposing 1,000+ scattered knowledge articles to teams throughout the department and replacing ad-hoc data pulls.
- Replaced a manual, hardcoded outage recovery process (100+ tickets) with an instant-failover interface for a ticketing system (Angular/Django) processing 2,000+ tickets/day, cut recovery time from hours to seconds and removed a recurring risk.
- Developed an automated pipeline that identifies customers impacted by service issues and dedupes event-driven operational data, generates resolution tickets and customer notifications, reducing manually-worked ticket volume by 60%.

Biamp Systems

Beaverton, OR

Software Engineering Co-Op

06/2023 - 12/2023

- Designed and built an LLM chatbot for internal engineering teams, including an automated ingestion pipeline that parses multiple document types (including C++ source) into a vector database (ChromaDB) organized for accurate, extensible retrieval.
- Developed custom retrieval workflows and backend APIs (data retrieval, session management, user settings) that routed queries to the correct document collection, giving engineers instant answers over docs/specs and code assistance, boosting productivity.

Digimarc Corporation

Beaverton, OR

Software Engineering Co-Op

06/2022 - 12/2022

- Developed a C++ ELF parser on Linux and integrated with existing SDK, enhancing software security.
- Optimized a Python-based image processing pipeline on AWS SQS/EC2 by adding per-job timeouts to prevent workers from hanging on oversized files and tightening autoscaling thresholds to stop overprovisioning, cutting costs by more than 50% and increasing throughput.

NASA Ames Research Center

Mountain View, CA

Software Developer Intern

06/2020 - 09/2021

- Developed the Celestial Mapping System: built on top of NASA WorldWind Java SDK, delivering 10 features in the first 3 months and taking a proof of concept to a functional application used to visualize and plan missions.
- Processed and integrated large geospatial datasets (GDAL/XML) to support real-time rendering and analysis of planetary and lunar surface data.
- Implemented a tessellation algorithm to reduce globe distortion and improve accuracy near the poles.

Education

Bachelor's in Software Engineering Technology

Oregon Institute of Technology - Klamath Falls, OR